# Religious Texts Platform

## Technical Documentation

*Version 0.1 | May 2026*

# 1. Project Overview

The Religious Texts Platform is a Vaadin-based web application for reading, comparing, and studying religious texts side-by-side. The system ingests texts from multiple sources into a BaseX XML database and presents them in a multi-column reader interface.

The platform is designed to support Bible translations, the Quran, Hadith collections, and associated commentaries in a single unified reader.

## 1.2 Technology Stack

| Layer | Technology |
|---|---|
| Frontend | Vaadin 24 (Java server-side rendering) |
| Backend | Spring Boot 3.2 |
| Text database | BaseX 10 (native XML database) — port 8984 |
| User database | MySQL 8.3 (users, comments, notes) — port 3306 |
| Build | Maven (multi-module) |
| Data ingestion | Spring Boot service on port 8091 |
| App server | Embedded Tomcat on port 8090 |
| Container | Docker (religioustext-basex, religioustext-mysql) |
| Bible source | Local git clone (wldeh/bible-api) + API.Bible |

## 1.3 Dual Database Architecture

The platform uses two databases for distinct purposes:

| Database | Technology | Purpose |
|---|---|---|
| religioustext-basex | BaseX 10 (XML) | All religious texts — Bible, Quran, Hadith. Read-heavy, complex XQuery, native XML storage. |
| religioustext-mysql | MySQL 8.3 | Users, comments, personal notes, sessions. Relational, write-heavy, Spring Security integration. |

Both databases run in Docker and are defined in docker/. BaseX is accessed via its REST API using XQuery over HTTP. MySQL is accessed via Spring Data JPA.

## 1.4 Module Structure

- app/ — Vaadin frontend + Spring Boot app (port 8090)
- ingestion/ — Ingestion service (port 8091)
- schema/ — XSD schema for the religioustext XML namespace
- docker/ — Docker configuration for BaseX (port 8984) and MySQL 8.3 (port 3306)

# 2. Data Architecture

## 2.1 XML Schema

All texts are stored in BaseX under the namespace http://religioustext.org/schema/1.0. The root element <text> carries translation-level metadata; books, chapters, and verses are nested within.

```
<text id="bible-kjv-1611" type="bible" translation="King James Version"
      abbreviation="KJV" bcp47Language="en" direction="ltr"
      license="Public Domain" year="1611" totalVerses="36820">
  <book name="Genesis" code="GEN" canonicalOrder="1" testament="OT">
    <chapter number="1" title="The Creation">
      <verse number="1" bookName="Genesis" chapterNumber="1">
        In the beginning God created...
      </verse>
    </chapter>
  </book>
</text>
```

## 2.2 Key Attributes

| Element | Attribute | Purpose |
|---|---|---|
| <text> | id | Document ID e.g. bible-kjv-1611 |
| <text> | totalVerses | Pre-computed verse count |
| <book> | code | Canonical API.Bible code (GEN, EXO... REV) |
| <book> | canonicalOrder | Sort order (1=Genesis, 66=Revelation) |
| <book> | testament | OT / NT / DC (Deuterocanonical) |
| <chapter> | title | Editorial section title (where available) |
| <verse> | number | Verse number within chapter |

## 2.3 Translation-Independent Verse Anchors

Each verse div in the UI is assigned an HTML anchor ID in the format verse-{CODE}-{chapter}-{verse}, for example verse-GEN-1-1. This is translation-independent — the same anchor ID is used across all open Bible columns, enabling commentary cross-references to highlight the same passage in every open translation simultaneously.

# 3. Ingestion Pipeline

## 3.1 Architecture

The ingestion service uses a producer/consumer pipeline with a configurable thread pool (default 5 crawler threads) feeding a single insert thread via a BlockingQueue. A poison pill sentinel terminates the insert thread cleanly after all books are processed.

## 3.2 Local Clone Ingestion (LocalIngestionService)

Reads Bible text from a local clone of the wldeh/bible-api repository. Chapter titles are optionally fetched from API.Bible (250ms rate limit between calls). Book metadata (canonical order, API code, testament) is resolved via a BOOK_META map keyed on folder name.

- Source path: /Volumes/VMs/data/bible-api-source/bibles/
- Supports English, Spanish, German, Greek, and other language folders
- BOOK_META covers Protestant canon + full DC/Apocrypha + Spanish (RVR09) folder names

## 3.3 API.Bible Ingestion (IngestionService)

Used for licensed translations (NIV, NASB20, NBLA) where the full text must be fetched from the API. Rate-limited; uses 3 licensed API.Bible slots.

## 3.4 Current Ingestion Status

| Translation | Language | Verses | Status |
|---|---|---|---|
| NIV 2011 | English | 30,752 | ✅ Complete — abbreviated book names, no code attr yet |
| KJV 1611 | English | 36,820 | ✅ Complete — inc. Apocrypha (80 books) |
| ASV 1901 | English | 31,102 | ✅ Complete |
| WEB | English | ~31K+ | ✅ Complete — inc. extra DC books |
| DRA 1899 | English Catholic | ~46K | ✅ Complete |
| RVR09 1909 | Spanish | 31,102 | ✅ Complete |
| NASB20 | English | — | ⏳ Not yet ingested |
| NBLA | Spanish | — | ⏳ Not yet ingested |

## 3.5 Known Issues

- NIV book names are abbreviated (Gen., Lev. etc.) — needs re-ingestion via API.Bible path to fix
- NIV <book> elements have no code= attribute — patched in all other translations via XQuery update
- Document naming inconsistency: NIV stored as bible-niv-2011.xml (with .xml suffix), others without

# 4. Application Layer

## 4.1 TextQueryService

Queries BaseX via its REST API using XQuery over HTTP. The executeQuery() method uses UriComponentsBuilder to avoid double URL-encoding of query parameters. All verse data is returned as pipe-delimited rows and parsed into VerseRef objects.

## 4.2 VerseRef Model

Immutable record carrying: sourceId, bookName, bookCode, chapterNumber, chapterTitle, verseNumber, content, globalCanonicalSeq, globalChronologicalSeq, globalNarrativeSeq, note.

## 4.3 DisplayOptions

| Mode | Description |
|---|---|
| Original | Scriptio continua — all caps, no spaces or punctuation, continuous block |
| Chapters | Chapter headings only, prose verse flow |
| Verses (default) | Chapter headings + superscript verse numbers |
| Titles | Chapters + verse numbers + editorial section titles |

## 4.4 ReaderView (UI)

Multi-column Vaadin layout. Each column is 33vw wide, horizontally scrollable. Columns are independent but sync by default — changing book/chapter in one synced column updates all other synced columns. Sync can be broken per column via the chain-link icon.

- Source selector: per-column ComboBox populated from BaseX listSources()
- Book/chapter nav: Select dropdowns with prev/next chapter buttons
- Sync toggle: LINK / UNLINK icon per column
- Display mode: per-column dropdown (Original / Chapters / Verses / Titles)

## 5. Notable Bug Fixes

| Bug | Fix |
| --- | --- |
| canonicalOrder was alphabetical (folder discovery order) | Use BOOK_META.canonicalOrder() not thread submission counter |
| InterruptedException noise on insert thread | Replaced interrupt() with poison pill (sentinel CrawledBook) |
| BaseXStore proc.waitFor() InterruptedException | Replaced with waitFor(30, TimeUnit.SECONDS) + destroyForcibly() |
| BOOK_META folder name mismatches (ecclesiasticus, manasseh, bel, esther(greek)) | Fixed keys to match actual folder names |
| RVR09 Spanish folder names not in BOOK_META | Added full Spanish mapping (génesis, éxodo... apocalipsis) |
| Double URL encoding in XQuery REST calls | Replaced URLEncoder + string URL with UriComponentsBuilder |
| book code= attribute missing from XML | Added to serialiseBook(); patched existing docs via XQuery update |
| Vaadin theme directory missing | Created app/src/main/frontend/themes/religious-texts/ |

## 6. Planned Features

### 6.1 Continuous Scroll

Infinite scroll with IntersectionObserver — loading next/previous chapters as the user scrolls. DOM window limited to previous + current + next book (~2,400 verse divs). Chapter selector updates as chapters cross viewport. Synced columns navigate (not pixel-scroll) to matching passage.

### 6.2 Quran & Hadith

Quran ingestion via fawazahmed0/quran-api. Hadith collections (Sahih al-Bukhari, Sahih Muslim, and the four Sunan) via a similar pipeline. Same XML schema with type="quran" and type="hadith".

### 6.3 Commentary & Cross-References

Commentary columns with clickable verse reference links. Links use the translation-independent anchor scheme (verse-GEN-1-1) to scroll all open Bible columns to the referenced passage simultaneously. If the referenced text is not open, offer to open a new column.

### 6.4 Bulk Translation Import

The local clone contains 200+ translations across dozens of languages. A future bulk import task will iterate all folders in /Volumes/VMs/data/bible-api-source/bibles/, auto-generate bible-sources.yml entries, and ingest everything. Requires language detection from folder name prefix and smarter BOOK_META fallback for non-Latin scripts.

### 6.5 User Features

- Spring Security login
- Personal notes column
- Verse highlighting and bookmarks
- Search across all ingested texts